

To calculate the Wilson Score Interval incorporating the maximum gap value θ :

1. ****Calculate Conditional Probabilities****: For each cluster size x , calculate the conditional probabilities $P(x + 1 | x)$ and $P(x | x - 1)$.
2. ****Calculate Differences****: Compute the differences $P(x + 1 | x) - P(x | x - 1)$ for each cluster size x .
3. ****Find Maximum Gap ()****: Identify the maximum value of these differences. This maximum value is your θ :

$$\theta = \max (P(x + 1 | x) - P(x | x - 1))$$

4. ****Adjust Proportion for Each Cluster Size****: For each cluster size x , adjust the observed proportion $\hat{p} = P(x + 1 | x)$ using θ :

$$\hat{p}_{\text{adjusted}} = \hat{p} - \theta$$

5. ****Calculate Wilson Score Interval****: Use the adjusted proportion $\hat{p}_{\text{adjusted}}$ in the Wilson Score Interval formula to calculate the confidence interval for each cluster size:

$$\hat{p}_{\text{adjusted}} \pm \frac{z_{\alpha/2}}{1 + \frac{z_{\alpha/2}^2}{n}} \sqrt{\frac{\hat{p}_{\text{adjusted}}(1 - \hat{p}_{\text{adjusted}})}{n} + \frac{z_{\alpha/2}^2}{4n^2}}$$

This adjusted confidence interval accounts for the most significant change in behavior observed, providing a more accurate reflection of the confidence in the observed behavior change.